

The General Gauss–Markov Model

Every chapter so far has assumed that the errors are uncorrelated and equally variable. That assumption is a statement about the *geometry* of observation space: it says that Euclidean distance is the right way to measure how far the data are from the model. When observations share a common shock, follow one another in time, or are measured with different precision, Euclidean distance is the wrong ruler. This chapter replaces $\sigma^2\mathbf{I}$ by a general $\sigma^2\mathbf{V}$: first with $\mathbf{V}$ known and invertible, where the theory of Parts II and III survives a change of coordinates; then in the two places where it does not survive, when $\mathbf{V}$ is singular and some functions of $\boldsymbol{\beta}$ become known exactly, and when $\mathbf{V}$ has to be estimated from the data that produced the fit.

What you need. [Chapter 6](#), especially [Section 6.9](#): projection in the $\mathbf{V}^{-1}$ inner product (Definition 6.9.1, Theorem 6.9.1), oblique projections (Proposition 6.9.4) and Kruskal’s theorem (Theorem 6.9.3). [Section 7.2](#) for the Gauss–Markov theorem (Theorem 7.2.1) and Aitken’s corollary (Corollary 7.2.3), which this chapter cites and does not reprove. From [Chapter 1](#): generalized inverses, square roots (Theorem 1.7.6) and the Woodbury identity (Theorem 1.6.5). From [Chapter 2](#): whitening (Proposition 2.2.4) and the support of a singular covariance (Theorem 2.2.3). Estimability ([Chapter 8](#)) and the F test ([Chapter 11](#)) are used throughout, and the distribution theory of [Section 31.1](#) is the quadratic-form theory of [Chapter 4](#), reached through Theorem 7.5.1 applied to the whitened model rather than cited here directly. [Section 19.2](#) and [Chapter 21](#) supply the diagnostic side: what a wrong covariance does (Theorem 19.2.2), how much efficiency it costs (Theorem 19.2.3), and the sandwich estimator (Theorem 21.4.2). [Section 31.1](#) also builds its interval on the t interval for an estimable function from [Chapter 12](#) (Theorem 12.1.1).

31.-1.1 Roadmap

- [Generalized least squares](#)
- [When ordinary least squares is best](#)
- [Singular covariance matrices](#)
- [Estimated covariance: feasible generalized least squares](#)
- [Summary and notes](#)